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CERTIFICATE

This is to certify that, **Mr. Abhay Chandrasen Jogiya** of **M.Sc. II Sem IV**, Roll No. **237252** has successfully completed the course **PHY-660-Research Project** entitled **A New $f(R)$ Model of Gravity** for partial completion of **M.Sc. Physics** degree of Fergusson College (Autonomous) affiliated to Savitribai Phule Pune University during the year 2024-2025.

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Declaration

I, **Abhay Chandrasen Jogiya**, hereby declare that the Project Report showcases my original work carried out at Fergusson College, Pune, under supervision of Mr. Gajanan Harale (Asst. Professor, Department of Physics, Fergusson College, Pune). I have not plagiarized any other source for the generation of this Project Report except where due reference or acknowledgement is made explicitly in the text.

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Abstract

At cosmological scales, gravity is governed by General Relativity (GR). However, to address the accelerating expansion of the universe without invoking a cosmological constant, we propose a modified gravitational framework by introducing corrections through an $f(R)$ extension of the Einstein-Hilbert action. This generalization leads to modified field equations that incorporate scalar degrees of freedom, offering a potential explanation for cosmic acceleration. By varying the action with respect to the metric, we derive the scalaron mass in the Jordan frame, providing insights into the dynamical behavior of the gravitational sector in this extended theory.

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Chapter 1

Introduction

At cosmological scales, gravity is described by the Theory of General Relativity (GR). Einstein's General Relativity remains the leading framework for understanding the geometry of spacetime and various gravitational phenomena. Despite its success, recent experimental findings have revealed deviations from its predictions. One of the most significant challenges in modern cosmology is the observed accelerated expansion of the universe, which GR struggles to fully explain. This issue was first identified approximately 22 years ago through observations of Type Ia supernovae [1, 2].

The limitations of GR, particularly on large scales, have raised fundamental questions regarding its completeness. The standard approach within GR requires invoking an unknown and invisible form of energy—dark energy—to account for roughly 76% of the universe's total energy content. Furthermore, GR is not renormalizable using conventional quantum field theory techniques, posing a fundamental theoretical challenge. To address these issues, various modifications of GR have been proposed. These alternative theories often introduce changes to the fundamental structure of spacetime, leading to deviations from GR predictions that can be tested observationally.

A straightforward extension of GR is the $f(R)$ theory of gravity, where the Ricci scalar R in the Einstein-Hilbert action is replaced by a more general func-

tion, $f(R)$. Over time, several $f(R)$ models have been proposed, some of which successfully address key limitations of GR. Among these, the Starobinsky model [3], the Hu-Sawicki model [4], and the Tsujikawa model [5], also known as exponential gravity, have gained significant attention for their ability to explain cosmic acceleration and structure formation.

1.0.1 Problem Statement

In the current, widely accepted Λ CDM model, the term Λ accounts for dark energy responsible for cosmic expansion. But the nature of Λ is not known because the nature of dark energy is not understood. Dark energy is said to be responsible for cosmic accelerated expansion in these models but the dark energy has not been detected or studied. Hence accepting Λ as entity responsible for cosmic accelerated expansion means to accept the unknown dark energy responsible for cosmic expansion without experimental evidence.

1.0.2 Project Objective

In this work, we introduce a new $f(R)$ gravity model to explore modifications of GR. Specifically, in an attempt to explain expansion of the universe without invoking Λ . In this work, we reach up to the part of deriving modified field equations for a new $f(R)$ theory of gravity and deriving scalaron mass in Jordan frame. Future work will proceed towards the end goal of explaining the expansion without invoking Λ .

Chapter 2

Literature Background

2.1 History

The development of gravitational theories has long been driven by the need to explain observed cosmic phenomena. Historically, Newtonian gravity provided an early framework, but its limitations became apparent with advancements in astrophysics. Einstein's General Relativity (GR) emerged as a revolutionary theory, accurately describing gravity as the curvature of spacetime. While GR has been successful in explaining many astrophysical and cosmological phenomena, new observations have revealed inconsistencies that challenge its completeness.

One of the major contemporary motivations for modifying GR comes from the discovery of cosmic acceleration, as evidenced by Type Ia supernovae observations. The standard approach to explaining this phenomenon involves introducing dark energy, often modeled through a cosmological constant (Λ). However, this approach raises theoretical concerns, including fine-tuning issues and discrepancies between observed and predicted vacuum energy densities. Additionally, GR is not renormalizable, making it incompatible with quantum field theory, which further motivates alternative formulations of gravity.

Among various modified gravity approaches, $f(R)$ theories represent a straightforward yet profound extension of GR. Instead of relying on the Einstein-Hilbert

action, where gravity is solely determined by the Ricci scalar R , $f(R)$ gravity generalizes the action by replacing R with an arbitrary function $f(R)$. These theories serve as useful toy models to explore deviations from GR and assess their implications on cosmology and astrophysics. Several models, such as the Starobinsky model and the Hu-Sawicki model, have been proposed to address the shortcomings of GR while preserving key observational successes.

By modifying gravity at large scales, $f(R)$ theories offer an alternative way to explain cosmic acceleration without invoking exotic dark energy components. This makes them an important area of research, helping to refine our understanding of gravitational interactions and potentially bridging the gap between classical and quantum gravity. [6]

2.2 Action

The action is a fundamental quantity that encapsulates the dynamics of a system. It is typically represented as an integral over time (or spacetime in field theories) and is given by:

$$S = \int \mathcal{L} d^4x \quad (2.1)$$

where $\mathcal{L}$ is the Lagrangian density, which describes the energy content and interactions of the system. The dynamics of the system are determined by applying the Principle of Least Action, which states that the physical path taken by a system is the one that makes the action stationary (typically a minimum). Mathematically, this means:

$$\delta S = 0 \quad (2.2)$$

where δS represents the variation of the action. This principle leads directly to the equations of motion governing the system. In classical mechanics, it gives

rise to Newton's laws, while in field theory, it leads to field equations such as the Einstein field equations in General Relativity.

2.3 Hilbert Space

Hilbert space is a fundamental concept in mathematical physics, providing the framework for quantum mechanics, functional analysis, and even aspects of field theory. It generalizes the notion of Euclidean space to infinite dimensions, allowing us to describe systems with infinitely many degrees of freedom. Reference from [7]

2.3.1 Definition of Hilbert Space

A **Hilbert space** $\mathcal{H}$ is a **complete inner product space**. This means it is a vector space over the field of complex numbers $\mathbb{C}$ (or real numbers $\mathbb{R}$) equipped with an inner product $\langle \cdot, \cdot \rangle$ that satisfies:

$$\langle a\psi + b\phi, \chi \rangle = a\langle \psi, \chi \rangle + b\langle \phi, \chi \rangle, \quad \forall a, b \in \mathbb{C}, \quad \psi, \phi, \chi \in \mathcal{H} \quad (2.3)$$

$$\langle \psi, \phi \rangle = \overline{\langle \phi, \psi \rangle} \quad (2.4)$$

$$\langle \psi, \psi \rangle \geq 0, \quad \text{and} \quad \langle \psi, \psi \rangle = 0 \quad \text{if and only if} \quad \psi = 0 \quad (2.5)$$

Additionally, a Hilbert space is **complete**, meaning that if a sequence of vectors $\{\psi_n\}$ is Cauchy (i.e., it converges in norm), then there exists a limit within the space:

$$\lim_{n \rightarrow \infty} \|\psi_n - \psi_m\| = 0 \Rightarrow \exists \psi \in \mathcal{H}, \quad \lim_{n \rightarrow \infty} \psi_n = \psi \quad (2.6)$$

where the **norm** induced by the inner product is:

$$\|\psi\| = \sqrt{\langle \psi, \psi \rangle} \quad (2.7)$$

2.3.2 Examples of Hilbert Spaces

1. Finite-Dimensional Hilbert Space: If $\mathcal{H}$ is **finite-dimensional**, it behaves like $\mathbb{C}^n$ with the usual Euclidean inner product:

$$\langle \mathbf{v}, \mathbf{w} \rangle = \sum_{i=1}^n v_i \overline{w_i} \quad (2.8)$$

2. Infinite-Dimensional Hilbert Space: Consider the space of **square-integrable functions** $L^2(\mathbb{R})$:

$$L^2(\mathbb{R}) = \left\{ \psi : \mathbb{R} \rightarrow \mathbb{C} \mid \int_{-\infty}^{\infty} |\psi(x)|^2 dx < \infty \right\} \quad (2.9)$$

The inner product is:

$$\langle \psi, \phi \rangle = \int_{-\infty}^{\infty} \psi^*(x) \phi(x) dx \quad (2.10)$$

2.3.3 Basis and Orthonormality in Hilbert Space

A **basis** of a Hilbert space is an infinite set of orthonormal vectors $\{e_n\}$ such that any vector ψ can be expanded as:

$$\psi = \sum_n c_n e_n, \quad \text{where} \quad c_n = \langle e_n, \psi \rangle \quad (2.11)$$

For example, in $L^2(\mathbb{R})$, the **Fourier basis** consists of functions:

$$e_n(x) = e^{inx}, \quad n \in \mathbb{Z} \quad (2.12)$$

which allows us to represent any function as a sum of sinusoidal components. A set $\{e_n\}$ is **orthonormal** if:

$$\langle e_m, e_n \rangle = \delta_{mn} \quad (2.13)$$

where δ_{mn} is the **Kronecker delta**.

2.3.4 Operators on Hilbert Space

Operators are linear maps on Hilbert space:

$$\hat{A} : \mathcal{H} \rightarrow \mathcal{H} \quad (2.14)$$

A special type of operator is the **Hermitian operator** (or **self-adjoint operator**), which satisfies:

$$\langle \psi, \hat{A}\phi \rangle = \langle \hat{A}\psi, \phi \rangle \quad (2.15)$$

Hermitian operators represent **observables** in quantum mechanics, such as position $\hat{x}$, momentum $\hat{p}$, and Hamiltonians $\hat{H}$.

2.4 Einstein-Hilbert Action

The Einstein-Hilbert action is given by:

$$S_{EH} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} R. \quad (2.16)$$

Here, $g = \det(g_{\mu\nu})$ is the determinant of the metric tensor, R is the Ricci scalar, and $\kappa = \frac{8\pi G}{c^4}$ is the gravitational coupling constant.

Varying the action with respect to the metric $g_{\mu\nu}$:

$$\delta S_{EH} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \delta R + \frac{1}{2\kappa} \int d^4x \delta(\sqrt{-g}) R. \quad (2.17)$$

Using the known variation of the determinant of the metric:

$$\delta(\sqrt{-g}) = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu}, \quad (2.18)$$

we obtain:

$$\delta S_{EH} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \delta R + \frac{1}{4\kappa} \int d^4x \sqrt{-g} g^{\mu\nu} R \delta g_{\mu\nu}.$$

Rearranging:

$$\delta S_{EH} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left(\delta R + \frac{1}{2} R g^{\mu\nu} \delta g_{\mu\nu} \right). \quad (2.19)$$

The Ricci scalar variation is given by the Palatini identity:

$$\delta R_{\mu\nu} = \nabla_\alpha \delta \Gamma_{\mu\nu}^\alpha - \nabla_\nu \delta \Gamma_{\mu\alpha}^\alpha. \quad (2.20)$$

Contracting with $g^{\mu\nu}$, we obtain:

$$\delta R = g^{\mu\nu} \delta R_{\mu\nu} = g^{\mu\nu} (\nabla_\alpha \delta \Gamma_{\mu\nu}^\alpha - \nabla_\nu \delta \Gamma_{\mu\alpha}^\alpha) \quad (2.21)$$

Substituting into the action (2.20):

$$\delta S_{EH} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} g^{\mu\nu} (\nabla_\alpha \delta \Gamma_{\mu\nu}^\alpha - \nabla_\nu \delta \Gamma_{\mu\alpha}^\alpha) + \frac{1}{4\kappa} \int d^4x \sqrt{-g} R g^{\mu\nu} \delta g_{\mu\nu}. \quad (2.22)$$

Using integration by parts, we rewrite:

$$\int d^4x \sqrt{-g} \nabla_\alpha V^\alpha = \int d^4x \nabla_\alpha (\sqrt{-g} V^\alpha) = \int d^4x \partial_\alpha (\sqrt{-g} V^\alpha),$$

which vanishes for a well-behaved boundary. This implies:

$$\int d^4x \sqrt{-g} g^{\mu\nu} \nabla_\alpha \delta \Gamma_{\mu\nu}^\alpha = \int d^4x \sqrt{-g} \nabla_\alpha (g^{\mu\nu} \delta \Gamma_{\mu\nu}^\alpha),$$

which is a total divergence and vanishes.

Thus, we obtain:

$$\delta S_{EH} = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left(R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} \right) \delta g^{\mu\nu}. \quad (2.23)$$

For the action principle $\delta S = 0$ to hold for arbitrary $\delta g_{\mu\nu}$, we require:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (2.24)$$

This is Einstein's field equation, describing the relation between the curvature of spacetime and the energy-momentum tensor $T_{\mu\nu}$.

If we modify the action by replacing R with a function $f(R)$, we get:

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} f(R) + S_m. \quad (2.25)$$

Here, S_m represents the matter action.

Varying the action with respect to the metric $g_{\mu\nu}$, we obtain:

$$\delta S = \frac{1}{2\kappa} \int d^4x \left[\delta(\sqrt{-g}) f(R) + \sqrt{-g} f'(R) \delta R \right]. \quad (2.26)$$

Using the variation of the determinant (2.18) we rewrite the action variation as:

$$\delta S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left[\frac{1}{2} f(R) g^{\mu\nu} \delta g_{\mu\nu} + f'(R) \delta R \right]. \quad (2.27)$$

Using the Palatini identity (2.20) contracting with $g^{\mu\nu}$, and integrating by parts, we obtain:

$$\delta S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left[f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \nabla^\alpha \nabla_\alpha - \nabla_\mu \nabla_\nu) f'(R) \right] \delta g^{\mu\nu}.$$

For arbitrary variations $\delta g_{\mu\nu}$, the modified field equations follow:

$$f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + (g_{\mu\nu} \nabla^\alpha \nabla_\alpha - \nabla_\mu \nabla_\nu) f'(R) = \frac{8\pi G}{c^4} T_{\mu\nu}. \quad (2.28)$$

Here, $f'(R) = \frac{df}{dR}$, indicating that the theory introduces extra degrees of freedom beyond General Relativity..

We need to design a viable form of $f(R)$ to get a suitable form of $f'(R)$. That $f'(R)$ is used in field equations to get modified theory of gravity which can be tested on various constraints and in various frames, which is the objective of this project.

Refer [6] and [7] for detailed calculations.

2.5 $f(R)$ Theories of Gravity

In both the metric and Palatini formalisms, $f(R)$ gravity can take the form of a Brans-Dicke theory representation [6]. This raises two natural questions: First, why should we consider the $f(R)$ formulation rather than just working with the Brans-Dicke one? Second, given that Brans-Dicke theory is well-understood, why should $f(R)$ gravity be regarded as unexplored or interesting?

The answer to the first question is fairly straightforward. There is no inherent reason to prefer one representation over the other, at least from the perspective of classical gravity. There may be specific applications where the $f(R)$ representation proves more convenient, and others where the Brans-Dicke representation is more suitable. It is worth noting that habit plays a role in shaping our preferences. Relativists often find the $f(R)$ representation more appealing due to its apparent geometric nature, while particle physicists tend to favor the Brans-Dicke representation. This preference can have theoretical implications. For instance, if $f(R)$ gravity is viewed as a stepping stone toward a more complex theory, the choice of representation may determine which generalizations are more straightforward to pursue [8].

As for whether $f(R)$ gravity is an unexplored theory or merely a subcase of Brans-Dicke theory, this is more of a practical question that merits a direct answer. It is true that scalar-tensor theories, particularly Brans-Dicke theory,

are well-studied and have been extensively applied in various domains, including cosmology. However, certain specific choices of the Brans-Dicke parameter, such as $\omega_0 = 0$ and $\omega_0 = -3/2$, are exceptional and have not been widely studied.

It is worth noting the following points:

1. For the case $\omega_0 = 0$, one could speculate that the absence of a kinetic term for the scalar field in the action made the theory less appealing, which may have hindered further investigation.
2. The choice $\omega_0 = -3/2$ results in a conformally invariant theory in the absence of the potential [9]. This initially led to the dismissal of Brans-Dicke theory, as the coupling with non-conformally invariant matter was not viable. However, if a potential is included, this issue is resolved, and the theory becomes more viable.

Additionally, most calculations in the literature that consider a general value for ω_0 exclude the case $\omega_0 = -3/2$, primarily for simplicity, as the combination $2\omega_0 + 3$ appears in the denominator. [8]

Chapter 3

Modeling A New f(R) Theory of Gravity

3.1 Action

In classical mechanics, the action S is defined as:

$$S = \int L dt \quad (3.1)$$

where L is is Lagrangian of the system, representing total energy of that system. The principle of least action states that physical trajectories are those for which the action is stationary (i.e., $\delta S = 0$) under small variations of the path.

In field theory, we generalize the Lagrangian to the Lagrangian density $\mathcal{L}$, which depends on the fields $\phi(x)$ and their spacetime derivatives $\partial_\mu(\phi)$. The action in field theory is given by:

$$S = \int d^4x \mathcal{L}(\phi, \partial_\mu(\phi)) \quad (3.2)$$

where $d^4x = dt d^3x$ represents the spacetime volume element and $\partial_\mu(\phi) = \frac{\partial \phi}{\partial x^\mu}$

Consider a small variation to the field: $\phi \rightarrow \phi + \delta\phi$. Then the variation in action is

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta(\partial_\mu \phi) \right] \quad (3.3)$$

Since

$$\delta(\partial_\mu \phi) = \partial_\mu(\delta\phi),$$

we rewrite:

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} \delta\phi + \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu(\delta\phi) \right] \quad (3.4)$$

Integrating by Parts

$$\int d^4x \frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \partial_\mu(\delta\phi) = \int d^4x \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \delta\phi \right) - \int d^4x \left[\partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial_\mu \phi)} \right) \delta\phi \right]$$

Dropping the surface term:

$$\delta S = \int d^4x \left[\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi)} \right) \right] \delta\phi$$

For arbitrary $\delta\phi$:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial(\partial^\mu \phi)} \right) = 0 \quad (3.5)$$

This is Euler-Lagrange Equation for Lagrangian density $\mathcal{L}$.

Now we derive the standard Lagrangian density for a real scalar field. To construct a physically meaningful Lagrangian density, we impose the following conditions:

1. **Lorentz Invariance:** The Lagrangian density must be invariant under Lorentz transformations. This ensures that the laws of physics remain unchanged for all inertial observers.
2. **Kinetic Term:** The Lagrangian should include terms that describe how the field varies in spacetime.
3. **Potential Term:** The field may have an interaction potential $V(\phi)$, which determines self-interactions or mass contributions.
4. **Linear in Second Derivatives:** To keep field equations second-order in derivatives (to maintain well-posed dynamics), we consider terms with at most **first-order derivatives** in ϕ .

To ensure Lorentz invariance, the scalar field Lagrangian takes the form:

$$\mathcal{L} = \frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi - V(\phi) \quad (3.6)$$

where $\eta^{\mu\nu}$ is the Minkowski metric.

The kinetic term $\frac{1}{2}\eta^{\mu\nu}\partial_\mu\phi\partial_\nu\phi$ ensures proper relativistic dynamics. The potential term $V(\phi)$ governs self-interactions and mass. This form respects Lorentz invariance, meaning the physics does not depend on the choice of inertial frame. Substituting (3.6) in (3.5) and computing derivatives:

$$\frac{\partial\mathcal{L}}{\partial\phi} = -\frac{dV}{d\phi}, \quad (3.7)$$

$$\frac{\partial\mathcal{L}}{\partial(\partial^\mu\phi)} = \eta^{\mu\nu}\partial_\nu\phi \quad (3.8)$$

and

$$\partial_\mu(\eta^{\mu\nu}\partial_\nu\phi) = \eta^{\mu\nu}\partial_\mu\partial_\nu\phi \quad (3.9)$$

Thus, the equation of motion is:

$$\eta^{\mu\nu}\partial_\mu\partial_\nu\phi = \frac{dV}{d\phi} \quad (3.10)$$

which can be written as:

$$\square\phi = \frac{dV}{d\phi}, \quad (3.11)$$

where $\square = \eta^{\mu\nu}\partial_\mu\partial_\nu$ is called the d'Alembertian operator.

This is the Klein-Gordon equation, which governs the dynamics of a free scalar field if $V(\phi)$ contains a mass term.

In General Relativity, the fundamental field is the metric tensor $g_{\mu\nu}$, and the Lagrangian density is given by:

$$\mathcal{L} = \frac{1}{2\kappa^2}R \quad (3.12)$$

where R is the Ricci scalar describing the spacetime curvature and then the Einstein-Hilbert action is:

$$S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} R \quad (3.13)$$

where $\sqrt{-g}$ accounts for the determinant of the metric term. Varying this action with respect to the metric $g_{\mu\nu}$ gives Einstein's field equations:

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4}T_{\mu\nu} \quad (3.14)$$

which describes how matter-energy $T_{\mu\nu}$ curves spacetime.

The full action of gravity and matter includes both the Einstein-Hilbert action and the matter action S_m :

$$S_{\text{total}} = S_{\text{EH}} + S_m$$

where S_m describes action for all matter fields:

$$S_m = \int d^4x \mathcal{L}_m \quad (3.15)$$

where $\mathcal{L}_m$ is matter Lagrangian Density.

Varying S_m with respect to $g_{\mu\nu}$:

$$\delta S_m = \int d^4x \delta(\sqrt{-g} \mathcal{L}_m)$$

Using product rule

$$\delta(\sqrt{-g} \mathcal{L}_m) = \mathcal{L}_m \delta(\sqrt{-g}) + \sqrt{-g} \delta \mathcal{L}_m$$

Using known identity for determinant of metric

$$\delta \sqrt{-g} = \frac{1}{2} \sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} \quad (3.16)$$

The variation becomes:

$$\delta S_m = \int d^4x \left(\frac{1}{2} \sqrt{-g} \mathcal{L}_m g^{\mu\nu} \delta g_{\mu\nu} + \sqrt{-g} \delta \mathcal{L}_m \right)$$

By definition, stress-energy tensor is:

$$T_{\mu\nu} = - \frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}} \quad (3.17)$$

Comparing the two, we get:

$$T_{\mu\nu} = \left(\frac{1}{2} \sqrt{-g} \mathcal{L}_m g^{\mu\nu} \delta g_{\mu\nu} + \sqrt{-g} \delta \mathcal{L}_m \right) \quad (3.18)$$

Simplifying gives:

$$T_{\mu\nu} = -\mathcal{L}_m g_{\mu\nu} + 2 \frac{\delta \mathcal{L}_m}{\delta g^{\mu\nu}} \quad (3.19)$$

3.2 Form of $f(R)$

In $f(R)$ gravity, the Ricci scalar R in the Einstein-Hilbert action is replaced by a general function $f(R)$, leading to the modified action:

$$S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} f(R) + S_m \quad (3.20)$$

Varying this action with respect to the metric leads to modified Einstein equations, which introduce additional terms that can explain cosmic acceleration without requiring dark energy. Therefore, we now design the $f(R)$ term which will be plugged into eq. (3.20) to get modified field equations. The form of $f(R)$ will be as follows:

$$f(R) = R + [\text{correction terms}]$$

We impose the following constraints for designing correction terms to ensure that the model is viable, consistent with known physics, and leads to testable predictions

1. **GR Recovery in High Curvature Regime** ($R \gg R_c$) The theory must recover GR at high curvature, i.e. $f(R) \rightarrow R + \text{small correction}$
2. **Cosmic Acceleration in Low Curvature Limit** ($R \ll R_c$) At low curvature $R \approx R_c \approx H_0^2$, the corrections must be non-trivial and drive late-time cosmic acceleration, mimicking dark energy.
3. **Stability Conditions** The corrections must be smooth, differentiable, and constructed so $f''(R) > 0$ across relevant R -values.
4. **No Singularities** The model must not introduce curvature singularities or discontinuities in derivatives.
5. **Observational Consistencies** The corrections must lead to predictions compatible with:

Cosmic microwave background (CMB)	Large-scale structure
Gravitational wave propagation	Solar system tests

For smooth transition between high and low curvature regimes, the correction terms must satisfy:

- $f_{\text{corr}}(R) \rightarrow 0$ **for** $R \gg R_c$

$f(R)$ modifications introduce new degrees of freedom and the field equations become of higher order. For the model to reduce to GR, the vanishing of correction terms at high curvature regime is a must.

- $f_{\text{corr}}(R) \approx \text{constant}$ **for** $R \gg R_c$

In GR, Λ accounts for dark energy which is attributed to cosmic accelerated expansion. Replacing Λ with effective term from $f(R)$ means the correction terms must mimic dark energy at low curvature.

The term R_c represents a characteristic curvature scale. It defines the transition between GR and modified gravity, defines the regimes where modifications become important. In General Relativity (GR), the Ricci scalar R varies depending on the physical system:

- High curvature: $R \sim 10^{54} \text{ m}^{-2}$ (early universe, black holes).
- Low curvature: $R \sim 10^{-52} \text{ m}^{-2}$ (current cosmic expansion).

The parameter R_c determines where the correction term starts influencing the evolution of the universe.

If R_c is too large, the modification only affects extreme low-curvature environments (e.g., the far future of the universe). If R_c is too small, the modification becomes relevant even at high curvature, potentially disrupting GR predictions in the early universe.

Thus, R_c is chosen so that modifications appear at the right cosmological epoch, such as at the onset of cosmic acceleration.

In standard GR with dark energy, the cosmological constant is roughly $R_c \sim 10^{-52} \text{ m}^{-2}$. In $f(R)$ gravity, R_c plays a similar role: it sets the scale at which deviations from GR become significant.

For large R - **High Curvature Regime**:

$$\frac{R_c^2}{R^2} \ll 1$$

say, $\frac{R_c^2}{R^2} = x$

Because the model should reduce to GR at higher curvature, the additional terms in $f(R)$ must naturally vanish, leaving only R behind.

One such naturally behaving function is the **inverse tangent**.

Taylor's expansion of $\tan^{-1}(x)$ at $x = 0$:

$$\tan^{-1}x = x - \frac{x^3}{2} + \frac{x^5}{5} - \dots$$

For $x \ll 1$, $\tan^{-1}x \sim x$ and higher terms vanish.

Using approximation:

$$\cot^{-1}x = \frac{\pi}{2} - \tan^{-1}x$$

$$\cot^{-1}x = \frac{\pi}{2} - x \quad \text{for } x \ll 1$$

$$\cot^{-1}\left(\frac{R_c^2}{R^2}\right) = \frac{\pi}{2} - \frac{R_c^2}{R^2} \quad (3.21)$$

Equation (3.21) represents the form of correction term for high curvature regime.

Introducing a free parameter α for controlling the correction terms and multiplying the term by R_c to set a transition limit, we get:

$$\boxed{f_{\text{corr}} = -\frac{\alpha}{\pi} R_c \left(\frac{\pi}{2} - \frac{R_c^2}{R^2} \right)} \quad (3.22)$$

Equation (3.22) is the final form of correction term for high curvature regime.

For small R - Low Curvature Regime:

$$x = \frac{R_c^2}{R^2} \gg 1$$

Now that we need the function to stabilize at low R , the function should grow as curvature decreases and avoid discontinuities, x becomes very large at $x \gg 1$.

Asymptotic analysis of the inverse tangent function:

$$\tan^{-1} x = \frac{\pi}{2} - \frac{1}{x} + \mathcal{O}(x^{-3}) \quad \text{for } x \gg 1.$$

The Big-O notation and is used in asymptotic analysis to describe the behavior of functions for small or large values of x

Using the identity:

$$\cot^{-1} x = \pi - \tan^{-1} x,$$

we get:

$$\cot^{-1} x = \pi - \left(\frac{\pi}{2} - \frac{1}{x} \right).$$

Simplifying:

$$\cot^{-1} x \approx \pi - \frac{1}{x}.$$

Substituting $x = \frac{R_c^2}{R^2}$:

$$\cot^{-1} \left(\frac{R_c^2}{R^2} \right) \approx \pi - \frac{R^2}{R_c^2} \tag{3.23}$$

The eq. (3.23) is the form the correction term takes at low curvature limit.

Again, using the free parameter α and R_c , we get:

$$\boxed{f_{\text{corr}} = -\frac{\alpha}{\pi} R_c \left(\pi - \frac{R^2}{R_c^2} \right)} \tag{3.24}$$

(3.24) is the final form of correction term in low curvature regime.

Combining the terms from eq. (3.21), (3.22), (3.23) & (3.24), we get correction term as:

$$\boxed{f_{\text{corr}} = -\frac{\alpha}{\pi} R_c \cot^{-1}\left(\frac{R_c^2}{R^2}\right)} \quad (3.25)$$

where

$$\boxed{\cot^{-1}(x) = \begin{cases} \frac{\pi}{2} - x, & x \ll 1 \quad (\text{Large } R) \\ \pi - \frac{1}{x}, & x \gg 1 \quad (\text{Small } R) \end{cases}} \quad (3.26)$$

and

$$x = \frac{R_c^2}{R^2}$$

The term (3.25) primarily modifies the gravity model at large curvature (high R). However, it does not provide a natural suppression mechanism at low curvature (small R). This can lead to two potential issues:

1. Residual Corrections at Small R :

The arctangent function does not decay rapidly at small R , meaning the modifications to GR persist even where we expect standard GR to hold (e.g., in the late universe or weak field regions).

This could lead to deviations from GR in solar system tests and other low-curvature astrophysical scenarios, which are tightly constrained by observations.

2. Asymptotic Behavior:

A well-constructed $f(R)$ model should reduce to GR in the limit $R \rightarrow 0$.

We need to devise a term that acts as a cutoff, ensuring that the correction vanishes smoothly as $R \rightarrow 0$, preserving the standard GR results in low-curvature environments.

If observational data indicate that deviations from GR should be large in high-curvature regimes (early universe) but suppressed in low-curvature regimes (late

universe), we need a function that naturally interpolates between these limits. [10, 11] Moreover, the theory should not introduce instabilities or ghost modes, which often arise if the corrections do not decay properly at low R . Additionally, to satisfy local gravity constraints, corrections to GR must become negligible in the weak field regime.[12]

We require that the correction term $f_{\text{corr}}(R)$ remains negligible at low curvature ($R \ll R_c$), meaning it must decay rapidly in this regime. A natural way to ensure this is an exponential suppression of the form:

$$f_{\text{corr}}(R) \sim \exp(-R/R_c) \quad (3.27)$$

Assuming the model will satisfy the following commonly found differential equation in $f(R)$ models: [6]

$$\frac{df(R)}{dR} = -\frac{1}{R_c} f(R) \quad (3.28)$$

Solving this ODE:

$$\frac{df}{dR} + \frac{1}{R_c} f = 0$$

Seperate variables:

$$\frac{df}{f} = -\frac{1}{R_c} dR$$

Integrate:

$$\int \frac{df}{f} = -\frac{1}{R_c} \int dR$$

$$\ln |f(R)| = -\frac{R}{R_c} + C$$

Solution is:

$$f(R) = C_1 e^{-R/R_c} \quad \text{where } C_1 \text{ is the constant of integration} \quad (3.29)$$

Eq. (3.29) gives direct motivation for the form of exponential suppression term.

At low curvature ($R \ll R_c$):

$$\exp(-R/R_c) \approx 1 - \frac{R}{R_c} + \frac{R^2}{2R_c^2} - \dots \Rightarrow 1 - \exp(-R/R_c) \approx \frac{R}{R_c} \quad (3.30)$$

So, the correction term behaves like a small linear deviation, suitable for late-time cosmic acceleration (small R).

At high curvature ($R \gg R_c$):

$$\exp(-R/R_c) \rightarrow 0 \Rightarrow 1 - \exp(-R/R_c) \rightarrow 1 \quad (3.31)$$

So, the function saturates and stops growing, meaning it doesn't interfere with early-time cosmology or high-curvature regimes like inflation or the early universe.

If we use just $f(R) \propto e^{-R/R_c}$, then:

At $R = 0$, $f(0) = 1$ (nonzero), which may not vanish where it physically should, like in Minkowski space, the behavior is not linear near $R = 0$, so it doesn't match General Relativity (GR) in the weak-field limit.

Using $f(R) \sim 1 - e^{-R/R_c}$ ensures $f(0) = 0$, recovering GR at low curvatures, provides a controlled, smooth deviation from GR at intermediate curvatures and naturally introduces a suppression scale R_c , which can be tuned to match observational data.

Hence, $1 - e^{-R/R_c}$ is the exponential suppression term we use for our model. Introducing a dimensionless parameter β to control the term and the characteristic curvature scale R_c , we get the exponential suppression part of the correction term as:

$$\boxed{f_{\text{corr}} = -\beta R_c \left(1 - e^{-R/R_c}\right)} \quad (3.32)$$

Eq. (3.25) and (3.32) combined make the total correction term to R and making our $f(R)$ as:

$$\boxed{f(R) = R + f_{\text{corr}}(R) = R - \frac{\alpha}{\pi} R_c \cot^{-1}\left(\frac{R_c^2}{R^2}\right) - \beta R_c \left(1 - e^{-R/R_c}\right)} \quad (3.33)$$

3.3 Modified Field Equations

Plugging this in eq. (3.20)

$$S_{\text{total}} = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} f(R) + S_m \quad (3.34)$$

$$= \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} \left\{ R - \frac{\alpha}{\pi} R_c \cot^{-1}\left(\frac{R_c^2}{R^2}\right) - \beta R_c \left(1 - e^{-R/R_c}\right) \right\} \\ + \left\{ \int d^4x \mathcal{L}_m \right\} \quad (3.35)$$

Varying the action with the metric $g_{\mu\nu}$ will give us modified field equations:

$$\delta S = \frac{1}{2\kappa^2} \int d^4x [\delta(\sqrt{-g}) f(R) + \sqrt{-g} \delta(f(R))] \quad (3.36)$$

Using

$$\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g}g_{\mu\nu}\delta g^{\mu\nu}$$

and

$$\delta f(R) = f'(R)\delta R$$

So the variation becomes:

$$\delta S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} \left[-\frac{1}{2}f(R)g_{\mu\nu} \delta g^{\mu\nu} + f'(R) \delta R \right] \quad (3.37)$$

Now we express δR in terms of $\delta g^{\mu\nu}$:

Ricci Tensor is:

$$R = g^{\mu\nu} R_{\mu\nu} \quad (3.38)$$

Varying with metric:

$$\delta R = \delta(g^{\mu\nu} R_{\mu\nu}) = \delta g^{\mu\nu} R_{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu} \quad (3.39)$$

Variation of Ricci tensor:

$$\delta R_{\mu\nu} = \nabla_\lambda \delta \Gamma_{\mu\nu}^\lambda - \nabla_\nu \delta \Gamma_{\mu\lambda}^\lambda$$

So:

$$g^{\mu\nu} \delta R_{\mu\nu} = \nabla_\lambda (g^{\mu\nu} \delta \Gamma_{\mu\nu}^\lambda) - \nabla_\nu (g^{\mu\nu} \delta \Gamma_{\mu\lambda}^\lambda)$$

This gives

$$g^{\mu\nu} \delta R_{\mu\nu} = \nabla_\mu \nabla_\nu \delta g^{\mu\nu} - \square(g_{\mu\nu} \delta g^{\mu\nu})$$

This identity follows from the variation of the Christoffel symbols:

$$\delta \Gamma_{\mu\nu}^\lambda = \frac{1}{2} g^{\lambda\sigma} (\nabla_\mu \delta g_{\sigma\nu} + \nabla_\nu \delta g_{\sigma\mu} - \nabla_\sigma \delta g_{\mu\nu})$$

Plug it in eq. (3.38)

$$\delta R = \delta g^{\mu\nu} R_{\mu\nu} + \nabla_\mu \nabla_\nu \delta g^{\mu\nu} - \square(g_{\mu\nu} \delta g^{\mu\nu}) \quad (3.40)$$

Plug this in (3.36) we get:

$$\begin{aligned} \delta S = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} \left[\left(f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} \right) \delta g^{\mu\nu} \right. \\ \left. + f'(R) (\nabla_\mu \nabla_\nu \delta g^{\mu\nu} - g_{\mu\nu} \square \delta g^{\mu\nu}) \right] \end{aligned} \quad (3.41)$$

Now, variation of matter action:

$$\delta S_m = \int d^4x \sqrt{-g} T_{\mu\nu} \delta g^{\mu\nu} \quad (3.42)$$

Then the total variation is:

$$\delta S_{\text{total}} = \frac{1}{2\kappa^2} \int d^4x \sqrt{-g} \left(f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} \right) \delta g^{\mu\nu} \quad (3.43)$$

$$\begin{aligned} + \frac{1}{2\kappa^2} \int d^4x f'(R) (\nabla_\mu \nabla_\nu \delta g^{\mu\nu} - g_{\mu\nu} \square \delta g^{\mu\nu}) \\ + \int d^4x \sqrt{-g} T_{\mu\nu} \delta g^{\mu\nu} \end{aligned} \quad (3.44)$$

We perform integration by parts on the second term (with covariant derivatives), discarding boundary terms:

$$\int d^4x f'(R) (\nabla_\mu \nabla_\nu \delta g^{\mu\nu} - g_{\mu\nu} \square \delta g^{\mu\nu}) \quad (3.45)$$

Integration of the first term:

$$\Rightarrow \int d^4x \sqrt{-g} f'(R) \nabla_\mu \nabla_\nu \delta g^{\mu\nu} \quad (3.46)$$

Applying identity for integration by parts in curved spacetime:

$$\begin{aligned} & \int d^4x \sqrt{-g} A^{\mu\nu} \nabla_\mu \nabla_\nu \phi \\ &= - \int d^4x \sqrt{-g} (\nabla_\mu \nabla_\nu A^{\mu\nu}) \phi + \text{boundary terms} \end{aligned} \quad (3.47)$$

where $f'(R) = A^{\mu\nu}$ and $\delta g^{\mu\nu} = \phi$

Since $f'(R)$ is a scalar, (3.45) simplifies to:

$$\int d^4x \sqrt{-g} f'(R) \nabla_\mu \nabla_\nu \delta g^{\mu\nu} = - \int d^4x \sqrt{-g} (\nabla_\mu \nabla_\nu f'(R)) \delta g^{\mu\nu} \quad (3.48)$$

Now integration of the second by-part term:

$$- \int d^4x \sqrt{-g} f'(R) g^{\mu\nu} \square \delta g_{\mu\nu} = - \int d^4x \sqrt{-g} f'(R) g^{\mu\nu} \nabla_\lambda \nabla^\lambda \delta g_{\mu\nu} \quad (3.49)$$

Now we apply integration by parts once to move one derivative from $g_{\mu\nu}$ onto $f'(R)$. Since ∇_λ acts on $g_{\mu\nu}$, we use the standard integration by parts in curved space:

$$\int d^4x \sqrt{-g} A^\lambda \nabla_\lambda B = - \int d^4x \sqrt{-g} (\nabla_\lambda A^\lambda) B \quad (\text{neglecting boundary terms})$$

Letting $A^\lambda = f'(R) g^{\mu\nu} \nabla^\lambda$ and $B = \delta g_{\mu\nu}$, we apply this to the term:

$$- \int d^4x \sqrt{-g} f'(R) g^{\mu\nu} \square \delta g_{\mu\nu} = - \int d^4x \sqrt{-g} f'(R) g^{\mu\nu} \nabla_\lambda \nabla^\lambda \delta g_{\mu\nu}$$

Now applying integration by parts to shift one derivative:

$$= \int d^4x \sqrt{-g} \nabla^\lambda (f'(R) g^{\mu\nu}) \nabla_\lambda \delta g_{\mu\nu}$$

Applying integration by parts again to shift the second derivative entirely off $\delta g_{\mu\nu}$:

$$= \int d^4x \sqrt{-g} g^{\mu\nu} \square f'(R) \delta g_{\mu\nu} \quad (3.50)$$

Now (3.47), (3.48) & (3.49) combined give (3.44) as:

$$\begin{aligned} & \int d^4x f'(R) (\nabla_\mu \nabla_\nu \delta g^{\mu\nu} - g_{\mu\nu} \square \delta g^{\mu\nu}) \\ &= \int d^4x \sqrt{-g} (-\nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R)) \delta g^{\mu\nu} \end{aligned} \quad (3.51)$$

And hence the total variation becomes:

$$\delta S_{\text{total}} = \int d^4x \sqrt{-g} \left[\frac{1}{2\kappa^2} (f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R)) - \frac{1}{2}T_{\mu\nu} \right] \delta g^{\mu\nu} \quad (3.52)$$

Setting $\delta S = 0$ for arbitrary $\delta g^{\mu\nu}$ should give us modified field equations:

So the integrand must vanish:

$$\frac{1}{2\kappa^2} \left(f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R) \right) - \frac{1}{2}T_{\mu\nu} = 0 \quad (3.53)$$

This gives:

$$\frac{1}{\kappa^2} \left(f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R) \right) = T_{\mu\nu} \quad (3.54)$$

$$\boxed{\Rightarrow f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R) = \kappa^2 T_{\mu\nu}} \quad (3.55)$$

(3.45) are modified field equations for our $f(R)$ model of gravity, where $f(R)$ is given by (3.33)

Chapter 4

Scaloron Mass

In the context of $f(R)$ gravity, the term scalaron refers to an additional scalar degree of freedom that emerges due to the modification of the Einstein-Hilbert action. This scalar field arises naturally when the Ricci scalar R in the gravitational action is replaced by a function $f(R)$. The scalaron can be interpreted as a particle with a specific mass, known as the scalaron mass.

The mass of the scalaron is crucial because it determines the range and strength of the fifth force mediated by the scalar field, influencing both cosmological dynamics and local gravitational experiments. In many $f(R)$ models, the scalaron mass depends on the local curvature or density, leading to mechanisms like the chameleon effect, where the scalaron's properties vary with the environment. [13, 14, 15]

To calculate scalaron mass in our model, we begin by taking trace of field equation (3.54) by contracting with $g^{\mu\nu}$

$$g^{\mu\nu} \left(f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R) \right) = \kappa T_{\mu\nu} g^{\mu\nu} \quad (4.1)$$

Breaking it down:

1. $g^{\mu\nu} (f'(R) R_{\mu\nu}) = f'(R) R$ Ricci Scalar
2. $g^{\mu\nu} (-\frac{1}{2} f(R) g_{\mu\nu}) = -\frac{1}{2} f(R) \times 4 = -2f(R)$

$$3. \ g^{\mu\nu}(-\nabla_\mu \nabla_\nu f'(R)) = -\square f'(R) \quad \text{by definition of } \square$$

$$4. \ g^{\mu\nu}(g_{\mu\nu}\square f'(R)) = 4 \square f'(R)$$

$$5. \ g^{\mu\nu}(\kappa^2 T_{\mu\nu} g^{\mu\nu}) = \kappa^2 T$$

Combining all together, (4.1) gives:

$$f'(R) R + 3 \square f'(R) - 2f(R) = \kappa^2 T \quad (4.2)$$

We define scalar field as $\phi = f(R)$, remembering that $f'(R) = \frac{df(R)}{dR}$.

Now calculating $f'(R)$ from (3.33):

$$\frac{d}{dR}f(R) = \frac{d}{dR}R - \frac{\alpha}{\pi} R_c \frac{d}{dR} \cot^{-1}\left(\frac{R_c^2}{R^2}\right) - \beta R_c \frac{d}{dR} \left(1 - e^{-R/R_c}\right) \quad (4.3)$$

Breaking it down:

$$1. \ \frac{d}{dR}R = 1$$

$$2. \ \frac{d}{dR} \cot^{-1}\left(\frac{R_c^2}{R^2}\right) = -\frac{1}{1+\frac{R_c^4}{R^4}} \left(-\frac{2R_c^2}{R^3}\right) = \frac{2R_c^2}{R^3+\frac{R_c^4}{R}}$$

$$3. \ \frac{d}{dR}[1 - e^{-R/R_c}] = 0 - \frac{d}{dR}[e^{-R/R_c}] = \frac{1}{R_c} e^{-R/R_c}$$

Combining this together, (4.3) becomes:

$$\frac{d}{dR}f(R) = f'(R) = 1 - \frac{\alpha}{\pi} \frac{2R_c^2}{R^3 + \frac{R_c^4}{R}} + \beta \exp(-R/R_c) \quad (4.4)$$

And since $\phi = f(R)$, using (4.4) in (4.2) gives the resultant equation which is viewed as scalar field equation of Klein-Gordan:

$$f'(R) R + 3 \square \phi - 2f(R) = \kappa^2 T \quad (4.5)$$

$$\Rightarrow \square \phi = \frac{1}{3}[2f(R) - R(\phi) + \kappa^2 T]$$

$$\frac{dV_{\text{eff}}}{d\phi} = \frac{1}{3}[2f(R) - R(\phi) + \kappa^2 T] \quad (4.6)$$

In absence of matter, $V_{\text{eff}} = V$. By stationary condition: $\frac{dV_{\text{eff}}}{d\phi} = 0$

Differentiating equation (4.6) gives:

First differentiating (4.4):

$$\frac{d}{dR} \left(-\frac{2\alpha R_c^2 R}{\pi(R^4 + R_c^4)} \right) = -\frac{(2\alpha R_c^2)(\pi(R^4 + R_c^4)) - (2\alpha R_c^2 R)(4\pi R^3)}{[\pi(R^4 + R_c^4)]^2}$$

Simplifying neumerator:

$$2\alpha R_c^2 \pi(R^4 + R_c^4) - 8\alpha R_c^2 \pi R^4 = 2\alpha R_c^2 \pi [(R^4 + R_c^4) - 4R^4] = 2\alpha R_c^2 \pi (-3R^4 + R_c^4)$$

$$\frac{d}{dR} \left(-\frac{2\alpha R_c^2 R}{\pi(R^4 + R_c^4)} \right) = \frac{2\alpha R_c^2 (3R^4 - R_c^4)}{\pi(R^4 + R_c^4)^2} \quad (4.7)$$

and

$$\frac{d}{dR} (\beta e^{-R/R_c}) = -\frac{\beta}{R_c} e^{-R/R_c} \quad (4.8)$$

Therefore,

$$f''(R) = \frac{2\alpha R_c^2 (3R^4 - R_c^4)}{\pi(R^4 + R_c^4)^2} - \frac{\beta}{R_c} e^{-R/R_c} \quad (4.9)$$

Hence, derivative of eq. (4.6) gives:

$$\frac{d^2 V_{\text{eff}}}{d\phi^2} = \frac{1}{3} \left(\frac{f'(R)}{f''(R)} - R \right) \quad (4.10)$$

At $\phi = \phi_0$:

$$\frac{d^2 V_{\text{eff}}}{d\phi_0^2} = \frac{1}{3} \left(\frac{f'(R_0)}{f''(R_0)} - R_0 \right) = m_\phi^2 \quad (4.11)$$

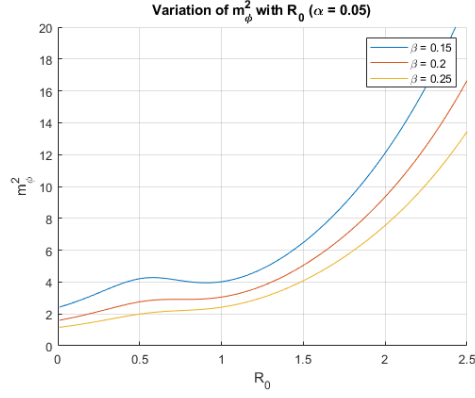
The term m_ϕ^2 represents the scalaron mass, R_0 is the background curvature corresponding to ϕ_0 . Plugging the values of $f'(R_0)$ & $f''(R_0)$:

$$m_\phi^2 = \left[\frac{R_c \exp(R/R_c) (\pi(R^4 + R_c^4)^2 - 8\alpha R^5 R_c^3) - \pi\beta(R + R_c)(R^4 + R_c^4)^2}{3\pi\beta(R^4 + R_c^4)^2 - 6\alpha R_c^4 \exp(R/R_c)(R_c^4 - 3R^4)} \right]_{R=R_0} \quad (4.12)$$

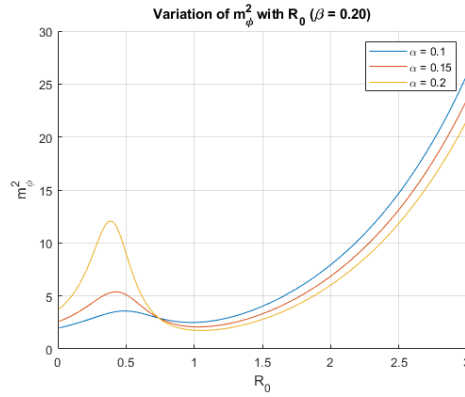
The eq. (4.12) gives the scalaron mass squared.

4.1 Variation of Sclaron Mass

From eq. (4.12) it is clear that m_ϕ^2 depends on α, β & R_c . If R_c is set to one, then the sclaron mass squared varies with background curvature R_0 as follows:



(a)



(b)

Figure 4.1: Mass square of the scalar field as a function of the background curvature for different sets of values of α and β with characteristic curvature constant $R_c = 1$ in arbitrary units.

4.2 Physical Interpretation

4.2.1 Dependence of Scalaron Mass on Curvature R_0

The mass squared of the scalar field, m_ϕ^2 , increases as the background curvature R_0 increases. This suggests that in high-curvature regimes, the scalaron becomes more massive, leading to a suppression of its effects at large R_0 . This behavior aligns with the expression we derived for the scalaron mass eq. (4.12).

4.2.2 Effect of α on m_ϕ^2 (Left Plot)

Keeping β constant and increasing α , we observe that the mass squared of the scalar field decreases slightly. This is expected from the correction term $f_{\text{corr}}(R)$ in the modified gravity function (3.33). The correction term modifies the effective dynamics of gravity and introduces additional screening effects.

4.2.3 Effect of β on m_ϕ^2 (Right Plot)

Keeping α fixed and increasing β , we see that the mass squared of the scalar field decreases. Since β controls an exponential suppression term in $f_{\text{corr}}(R)$, it plays a role in stabilizing the scalaron and altering its effective mass.

4.3 Physical Significance

These plots suggest that the scalaron mass m_ϕ is strongly dependent on both the background curvature R_0 and the correction parameters α and β . The dependence on α and β could indicate a transition between different gravitational regimes, similar to those found in modified gravity models such as Starobinsky gravity [3].

If $m_\phi^2 < 0$ for some parameter values, it could indicate a tachyonic instability, leading to undesired behavior in cosmological models.

In the context of cosmology, a large m_ϕ^2 means the scalar degree of freedom decouples quickly, making deviations from General Relativity less observable.

In the context of astrophysics, the variations in m_ϕ^2 could have implications for the formation of structures and modifications to the standard Λ CDM model.

Chapter 5

Results and Conclusion

We derive a new form of $f(R)$ given by:

$$f(R) = R + f_{\text{corr}}(R) = R - \frac{\alpha}{\pi} R_c \cot^{-1}\left(\frac{R_c^2}{R^2}\right) - \beta R_c \left(1 - e^{-R/R_c}\right)$$

And we vary the modified action under stationary conditions to find modified field equations for gravity:

$$\Rightarrow f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R) + g_{\mu\nu} \square f'(R) = \kappa^2 T_{\mu\nu}$$

and we find scalaron mass term for our modified gravity as:

$$m_\phi^2 = \left[\frac{R_c \exp(R/R_c) (\pi(R^4 + R_c^4)^2 - 8\alpha R^5 R_c^3) - \pi\beta(R + R_c)(R^4 + R_c^4)^2}{3\pi\beta(R^4 + R_c^4)^2 - 6\alpha R_c^4 \exp(R/R_c)(R_c^4 - 3R^4)} \right]_{R=R_0}$$

and study its variation as function of background curvature. All of the above calculations are done in Jordan frame, which is a representation of a gravitational theory in which the metric tensor $g_{\mu\nu}$ is the one that directly couples to matter fields. This means that the equations of motion for test particles follow the standard geodesic equation, making it the natural frame for describing physical measurements like distances and time intervals.

Chapter 6

Future Scope

This work includes deriving modified gravitational field equations and corresponding scalaron mass in Jordan frame. This work can be extended to study the modified gravity in Einstein frame.

The model of gravity devised here can be tested under solar system constraints, can be passed through chameleon mechanisms and it can be used to study properties of various cosmological phenomena, such as gravitational waves, under the modified field equations.

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